The Twist: inflation, matter and dark energy from one supergravity field

A pre-registered exploratory study

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Abstract

We study whether one field can account for inflation, the matter–antimatter asymmetry and today's dark energy. The field space of the "Spiral Twist" model, a dilaton plus the Barbero–Immirzi angle with curvature $K = -2/3$, is identical to that of no-scale supergravity, $K = -3 \ln(T + \bar{T})$. The inflaton is Re T and the Immirzi angle is its partner Im T. The angle couples to the fermion axial current with a strength fixed by the Nieh–Yan term, $c_A = \sqrt{(3/8)}$; it is not a free parameter.

In a potential ordered as plateau → cliff → tail, the plateau gives an $\alpha = 1$ attractor with $n_s \approx 0.963$ and $r \approx 0.004$. A small non-perturbative term ($\delta \approx 10^{-4}$–10^{-5}) kicks the angle at the cliff, and the curved field space then boosts its speed during kination. Through the neutrino-mass (Weinberg) operator and sphalerons, the angle's rotation produces the observed baryon asymmetry, $n_B/s = 8.7 \times 10^{-11}$. The tail has slope $\sqrt{(2/3)}$, set by $\alpha = 1$.

Against late-time data (supernovae, CMB distance priors, DESI and SDSS BAO), the theory's own late-time prediction, a dark-energy tail of slope $\sqrt{(2/3)}$ with no dark-matter coupling and no parameter beyond ΛCDM's, passes all three pre-registered prediction tests: $\Delta\chi^2 = -3.4$ with DESI, and it predicts held-out DESI (−0.9) and SDSS (−0.4) data better than ΛCDM. The preference is modest and not blind. Versions with a fitted coupling fit better but do not predict; the one-parameter flagship (slope 1) predicts withheld DESI data worse than ΛCDM. We also report negative results: a simple vacuum makes the angle oscillate and erase the asymmetry, a smooth tail never ends inflation, and the gravitino cannot be the dark matter. The matter mechanism is lepto-axiogenesis, already published; what the Twist adds is a specific single-field embedding with a derived coupling. All data tests were written and locked before they ran. Frozen predictions: $r = 0.0037$–0.0040, $n_s \approx 0.963$–0.965, no cosmic birefringence from this field, and a dark-energy tail of slope $\sqrt{(2/3)}$ with $\beta = 0$.

1. Introduction and scope

Three facts about the universe usually get three separate explanations: an early period of inflation, a small excess of matter over antimatter, and a late acceleration attributed to dark energy. This paper asks whether one field, the Spiral Twist, can account for all three. The question is put to explicit calculations and to pre-registered tests against data.

What is claimed. The Twist's two-field geometry is the field space of no-scale supergravity at $\alpha = 1$. In that geometry, one explicit model reproduces the observed inflation spectrum, the observed baryon asymmetry and a dark-energy tail, with the angle's coupling to matter derived rather than chosen. The model makes specific predictions that upcoming experiments can falsify.

What is not claimed. The late-time fits do not establish evidence against ΛCDM: the flagship's preference over ΛCDM is 0.2–1.0σ after accounting for the number of tests run. The matter mechanism is not new (Section 8). Parts of the model, the kick term and the reheating, are put in rather than derived (Section 9).

How the work was done. Every data test has a written rule, fixed with a SHA-256 hash before its numbers were computed. Rules are never edited afterwards; deviations go in a separate log written before any rerun, and each test is reported once, whatever it shows. Theory explorations (Sections 2–4 and 6) are not pre-registered and are labelled exploratory. The author is an independent researcher. The calculations, code and drafting were done with substantial assistance from an AI system (Claude, Anthropic), working under the author's direction and the rules above.

2. The model

Field space. The Twist has a dilaton φ and an angle θ (the Barbero–Immirzi field), with kinetic metric

(1) $\quad ds^2 = d\varphi^2 + \exp(\lambda\varphi)\, d\theta^2, \quad \lambda = -2\sqrt{(2/3)}$

in reduced Planck units. Its curvature is $K = -\lambda^2/4 = -2/3$, the $\alpha = 1$ attractor geometry. Writing $T = x + iy$ with $x = \exp(\sqrt{(2/3)}\varphi)$, the Kähler potential $K = -3 \ln(T + \bar{T})$ gives exactly this metric (the y direction carries $\exp(-2\sqrt{(2/3)}\varphi)$; checked symbolically). The inflaton is Re T and the Immirzi angle is Im T: one complex supergravity field. This identification is the Twist's starting point, not a result of this paper: the Barbero–Immirzi chiral superfield of Gates, Ketov & Yunes (2009) has $K_{\Phi\bar{\Phi}} = 3/(\Phi + \bar{\Phi})^2$, from which the Twist's slope $\lambda = -2\sqrt{(2/3)}$ was derived earlier in this project. A constant superpotential gives the angle no potential at all, so a further sector is needed. The potential below supplies one.

Coupling to matter. With fermions present, the Nieh–Yan term makes the angle couple to the fermion axial current with strength $c_A = \sqrt{(3/8)}$ (Mercuri 2009, Eq. 11; Calcagni & Mercuri 2009, Eqs. 20–22):

(2) $\mathcal{L} \supset (c_A / M_P)\, \partial_\mu\theta\, J_5{}^\mu, \quad c_A = \sqrt{(3/8)}$

A rolling angle therefore acts as a chemical potential $\mu_A = c_A\, \dot\theta/M_P$ that tilts matter over antimatter.

Potential. We use the stabiliser construction $K = -3\ln(T + \bar T - S\bar S)$, $W = \sqrt{3}\, m\, S\, F(T)$, which gives $V = m^2|F(T)|^2/(4x^2)$ at $S = 0$. The working model is

(3) $F(T) = [\, T/\sqrt{(1 + T/\mu)} + \delta \exp(i\gamma)\, T \exp(-a\, m_k/T) / (1 + T/\mu)\,] \times \exp(-n(1 - e^{-\pi T/4}))$

Piece	Role	Values used
$T/\sqrt{(1 + T/\mu)}$	plateau at small x; tail $V \propto \exp(-\sqrt{(2/3)}\varphi)$ for $x \gg \mu$	$\mu = 10^4$
$\exp(-n(1 - e^{-\pi T/4}))$	cliff: ends inflation, drops the energy by $\exp(-2n)$	$m_2 = 1$; n = 10 (numerics), 125 (realistic)
$\delta \exp(i\gamma)\, T \exp(-a\, m_k/T)/(1 + T/\mu)$	kick: moves the angle's minimum at the cliff; γ sets matter vs antimatter	$m_k = m_2/2n$, $a = 1$, $\gamma = \pi/2$, $\delta \approx 10^{-4}$–10^{-5}

The order matters: a stretch of tail before the cliff inflates on its own ($\varepsilon = 1/3$), so the tail must start after the cliff (Section 6). The plateau sits at small T, which is the mirror image ($T \to 1/T$) of the usual placement; the reason is also given in Section 6.

Cliff factor (revised 13 Nov 2026). Earlier versions used $\exp(-nT/(T + m_2))$. Its leftover slope on the tail dies only like 1/x, and at the computed reheating (heat share ≈ 0.13%) it makes today's dark energy too stiff: a locked test of that full model (TAIL-F) gives $\Delta\chi^2 = +29.2$ against ΛCDM. The factor $\exp(-n(1 - e^{-\pi T/4}))$ has the same linear form on the plateau — inflation is unchanged (n_s 0.9642, r 0.0038 at instant reheating) — while its leftover slope dies like $e^{-\pi x/4}$, so the tail has slope $\sqrt{(2/3)}$ at any heat share ($w_0 - 0.90$, $w_a - 0.145$). $e^{-\pi T/4} = e^{-2\pi T/8}$ is the exponent of an SU(8) condensate, and the factor is periodic in the angle with period 8. This revision was made after the TAIL-F result and is a choice, not a derivation; only new data test it. The tail energy requires n ≈ 124 (chosen). Keeping the helper field S at zero needs the standard term $\zeta(S\bar S)^2$ inside K with $\zeta \gtrsim 3\times10^4$.

3. Inflation

The α = 1 plateau gives n_s ≈ 0.963–0.965 and r ≈ 0.0034–0.0040, with no parameter fitted. The scalar amplitude fixes the scale: V*^(1/4) ≈ 7.8 × 10^{15} GeV and H_inf ≈ 1.5 × 10^{13} GeV.

Calculation	N*	n_s	r	Status
E-model plateau, instantaneous reheating, exact slow roll (TWIST-INFLATION)	55.6	0.9653	0.0034	pre-registered, computed
Same, reheating anywhere from 10^{13} GeV (matter's minimum) to 2.6 × 10^{15} GeV (instant)	53.7–55.6	0.9642–0.9653	0.0034–0.0037	exploratory
Plateau → cliff → tail model of Section 2, cliff depth n = 10 / 125	55.6	0.9631 / 0.9635	0.0040 / 0.0039	exploratory

Against current measurements, n_s ≈ 0.963–0.965 sits within 0.5σ of Planck 2018 (0.9649 ± 0.0042), 1.0–1.8σ below Planck+ACT+SPT (0.9684 ± 0.0030), and 2.6–3.3σ below Planck+ACT+DESI (0.9743 ± 0.0034). The tensor ratio is far below today's bound (r < 0.036). The decisive test is a future r measurement with σ(r) ⩽ 0.002: the pre-registered reading is **SUPPORTED** if r is detected and consistent with 0.0037–0.0040 within 2σ, and **RULED OUT** if r < 0.002 or r > 0.008 at 95%. The full calculation's r = 0.0034 sits just under that frozen leading-order band; the band was not changed.

No matter-density patches. During inflation the angle's mass along its valley is m/H ≈ 2 in the stabiliser model and 15–45 in the moving-minimum variant of Section 4. Both exceed 3/2, so its fluctuations decay and leave no baryon isocurvature at leading order. Adding the kick term changes neither n_s nor r (0.9648 and 0.0036 in every case tested).

Can n_s move within the theory? With α = 1 fixed, we computed every route (exploratory). A stiff kination era after the cliff adds at most +0.0007 while reheating is hot enough for matter formation; second-order slow roll adds ~10^{-5}; the angle has no effect. Only cutting inflation early (a cliff 12–19 e-folds up the plateau) reaches 0.971–0.974, and then r falls to 0.0024–0.0020. The model's own cliff settings cannot do this: the position m2 only shifts the field, and softening the cliff lowers n_s (to 0.9629 at the steepness threshold n ≈ 3.46). We therefore keep the frozen prediction and record an early cliff only as an unlocked hypothesis; the locked NS-PURE test (Section 5) asks whether the late-time Twist lowers the n_s preferred by the CMB.

Other inflation observables. At the model's own N* = 57.6 (including kination): running $\alpha_s = -6.0\times10^{-4}$, tensor tilt $n_t = -r/8 = -4.4\times10^{-4}$ (single-field consistency), squeezed non-Gaussianity $f_{NL} \approx 0.014$, inflation scale $V^{1/4} \approx 7.9\times10^{15}$ GeV and $H^* \approx 1.5\times10^{13}$ GeV. The hidden SU(8) confines near 10^{14} GeV and its glueballs decay through Planck-suppressed operators in $\sim10^{-3}$ s, before nucleosynthesis.

4. Matter formation

The angle makes the observed matter if it moves at $r \equiv \dot\theta/(H M_P) \approx 0.02$ when the neutrino-mass processes are active. In the model of Section 2 that happens for $\delta \approx 10^{-4}$–10^{-5}, depending on how efficient reheating is.

The chain. The rolling angle acts as a chemical potential μ_A for fermion chirality. Fast gauge, Yukawa and strong-sphaleron processes spread it to leptons, with $\mu_\ell/\mu_A = 0.5227$ in the set of processes fast above 10^{12} GeV. The lepton-number-violating Weinberg operator (the origin of neutrino masses) relaxes lepton number toward

(4) $$Y_{eq} = -(45 / 2\pi^2 g^*)\,(\mu_\ell/\mu_A)\,(\mu_A/T), \quad \Gamma_W = 3T^3\,\Sigma m_\nu^2 / (4\pi^3 v^4)$$

and electroweak sphalerons convert 28/79 of it into baryons. This is the lepto-axiogenesis mechanism (Section 8). Neutrino masses are taken from oscillation data (normal ordering, lightest 0); they set the temperature where the process freezes out, about 2×10^{13} GeV.

Pre-registered check (MATTER). For a freely rolling angle ($\dot\theta \propto a^{-3}$) in a radiation era, the required speed is $r = 0.022$ at $T_{RH} = 10^{13}$ GeV, 0.005 at 3×10^{13}, 0.0035 at 10^{14} and 0.030 at 10^{15} GeV. The rule's reading is **VIABLE**. The angle's energy share stays below 0.015%.

Exploratory: the explicit model. At the cliff the kick term moves the angle's minimum and sets it rolling. During the kination that follows, the inflaton runs toward larger T, the angle's metric factor shrinks, and the curved field space boosts its speed. Radiation from reheating (instant preheating, efficiency f) then takes over. Feeding the model's own angle-speed history and expansion rate into the same rate equation:

Reheating efficiency f	δ for $n_B/s = 8.7 \times 10^{-11}$	Angle's peak energy share
0.99	1.5×10^{-4}	$\sim10^{-7}$
0.1	4.0×10^{-5}	$< 10^{-5}$
0.01	1.8×10^{-5}	$< 10^{-5}$

The mechanism is efficient: larger δ overproduces matter ($\delta = 10^{-3}$ gives 7× the observed amount at f = 0.99). The angle reverses direction twice, late, which lowers the yield by about 6%. The sign of the asymmetry follows the phase γ and is not predicted. Instant-preheating efficiency scales as f ≈ g^(5/2) for a coupling g, so the whole range above is plausible.

Where the kick comes from (exploratory). A kick derived from the dilaton's own sector fails. An instanton interfering with supersymmetry breaking (the angle potential ($A W_0 W_c/2x^2$) $\exp(-Ax)\cos(Ay)$, derived symbolically) carries a factor $\exp(K) \propto x^{-3}$. Inflation and dark energy sit at opposite ends of the field range, so that factor makes the term either swamp one end (by up to 10^{25}) or, if it shares the cliff, become far too weak (best window ~10^{-12} against ~10^{-4} needed). This is the known clash between moduli stabilisation and quintessence. A thermal kick is short by ~10^{164}. What works is a separate hidden field σ sharing a cosine with the Immirzi angle at the Nieh–Yan period. Holomorphy makes that cosine's strength scale as $\exp(-A\,\mathrm{Re}\,T)$, so it must act early, before the dilaton rolls far: a hidden mass of ~1.3×10^{11} GeV releases it at ~3×10^{14} GeV, where Re T ≈ 1. In the full model the observed asymmetry then needs the cosine to carry ~1.3×10^{-4} of the energy at release (10^{-4} → 0.77×, 3×10^{-4} → 2.3×, 10^{-3} → 7.5×), about 3×10^{-3} of the hidden field's own energy. That ratio need not be put in: if the cosine starts at its natural strength and is weakened only by holomorphy's $\exp(-A\,\mathrm{Re}\,T)$ as the dilaton rolls, the observed asymmetry comes out for a hidden mass of ~8×10^{9} GeV (3×10^{9} → 0.15×, 10^{10} → 1.6×, 3×10^{10} → 13×). The matter step then has no unexplained small number; its one free input, the hidden mass, is fixed by the observed amount of matter. Isocurvature ($\delta\sigma/\sigma \approx 10^{-5}$) is safe; the angle's energy share stays $\leqslant 3 \times 10^{-6}$, and the hidden field's own share peaks at ~5%, so it must decay before element formation (easy at 10^{11} GeV). The result holds in the frame the base theory's own conventions select, and drops 400× in the other frame. σ must decay, so it is not the dark matter.

Decay, derived reheating and the hidden sector's origin (exploratory). With the realistic cliff (n = 125) the hidden field would otherwise hold over half the energy; a decay coupling y = 0.1 removes it (share at the end 3×10^{-7}) at a cost of only an entropy dilution of 1.2–1.5. Reheating need not be put in by hand: instant preheating with a coupling g ≈ 0.1 gives an efficiency f ≈ 0.15–1.5% and $T_{RH} \approx 3$–5×10^{14} GeV from the model's own inflaton speed ($H_{end} = 5.2\times10^{12}$ GeV). At that derived f the matter yield is 1.1–4.4× observed. The yield falls as the hidden mass rises (8.8× at 2.9×10^{9} GeV, 4.3× at 6×10^{9} GeV, 2.5× at 1.2×10^{10} GeV; a fit gives $\propto m_\sigma^{-0.89}$), so the observed amount needs $m_\sigma \approx 3\times10^{10}$ GeV; an earlier estimate of 3–6×10^{9} GeV, based on the n = 10 scaling, was withdrawn after a pre-stated check failed. The required cosine, of natural height and suppressed only by $\exp(-A\,\mathrm{Re}\,T)$, is the standard cross term from gaugino condensation of a hidden gauge group with gauge

kinetic function f = T [Li, Li & Nanopoulos, arXiv:1407.1819]; for SU(N), a = 2π/N, so the Nieh–Yan period 8 (A = π/4) corresponds to a hidden SU(8). The term is negligible during inflation (at most 10^{-6} of the inflaton slope).

5. The late universe

The theory's own prediction for today's universe, a dark-energy tail of slope $\sqrt{(2/3)}$ with dark matter uncoupled (β = 0), fits and predicts the late-time data better than ΛCDM with no extra parameter (PURE-TWIST: **PREFERRED**). Earlier versions with fitted parameters fit well but did not predict.

Models. In the late universe the Twist field is a scalar with potential $V \propto \exp(-\kappa\varphi)$, optionally coupled to dark matter with strength β. The flagship (FLAG-1) fixes κ = 1 and fits β ≈ 0.041. The supergravity model predicts $\kappa = \sqrt{(2/3)} = 0.8165$ (the α = 1 tail slope, matching Dimopoulos & Owen's $2/\sqrt{(6\alpha)}$) and β = 0 (Section 6).

Tests. Each version is fitted three ways: with DESI DR2 BAO; with no BAO, after which it predicts DESI and SDSS; and with SDSS BAO in place of DESI. A version **predicts** if both held-out predictions beat ΛCDM and its SDSS fit is no worse than ΛCDM by more than 1. It is **preferred** if it also improves the DESI fit after the AIC penalty of 2 per extra parameter. All numbers are differences from ΛCDM (negative is better):

Version	Extra parameters	ΔAIC, fit with DESI	Predicting DESI	Predicting SDSS	Fit to SDSS	Locked reading
PURE-TWIST (κ = $\sqrt{(2/3)}$, β = 0)	0	**−3.37**	**−0.89**	**−0.41**	**−2.67**	**PREFERRED**
KAPPA-ALPHA1 (κ = $\sqrt{(2/3)}$, β fitted)	1	−4.30	−6.72	+0.90	−3.04	not predictive
FLAG-1 (κ = 1, β fitted)	1	−3.32	+10.88	+10.38	−1.20	not predictive
SPIRAL (κ, β both fitted)	2	−2.33	−6.27	+0.39	−3.27	not predictive
ON2 (coupling switches on late)	1	+0.53	−1.90	−0.37	−0.78	predicts, not preferred
SPIRAL0 (κ = 1, β = 0)	0	+1.11	+0.62	−0.36	+1.02	not predictive
w_0w_a (for comparison)	2	−2.21	−3.87	−0.17	−3.27	—

Reading. With β = 0, the slope the geometry predicts (√(2/3)) beats the flagship's slope of 1 on every measure (Δχ² −4.49 with DESI against SPIRAL0). Fitting β improves the DESI fit but spoils the predictions: KAPPA-ALPHA1 and FLAG-1 fit DESI by drawing on DESI itself. The flagship, refitted without DESI, predicts DESI worse than ΛCDM by 12.5 (DESI-HOLDOUT Test A). Its edge sits mostly in DESI's z ≈ 0.71 bin (M1), where DESI and SDSS differ by about 1.8σ in transverse distance, though the two surveys agree overall (p = 0.62, M3).

Caveats. PURE-TWIST's preference is modest: Δχ² = −3.4 is a likelihood ratio of about 5. It is not blind, since related fits (SPIRAL0, KAPPA-ALPHA1) were known when its rule was written, and it is one of many tests run on these data. The likelihood is compressed (distance priors, not the full CMB). The reading is a reason to test the √(2/3), β = 0 prediction on new data (DESI DR3, Euclid), not evidence for it yet.

The real supergravity tail. The model's tail is $V \propto \exp(-2n)/(1 + x/\mu)$, flat for $x \ll \mu$ and exponential with slope √(2/3) only for $x \gg \mu$. Following the field from the end of inflation until it freezes gives $x_f \approx 2$–300 depending on the reheating share, so the slope seen today is $\sqrt{2/3}\cdot x_f/(x_f + \mu)$. For $\mu \lesssim 1$ this is 97–100% of √(2/3), i.e. PURE-TWIST; for the earlier choice $\mu = 10^4$ it is ≈ 0.3–3%, i.e. almost a cosmological constant. We adopt $\mu \lesssim 1$, noting that the expansion data already seen favour it, so its test must use new data (WATCH-PURE-DR3). The cliff depth then sets the dark-energy scale: $\exp(-2n)\, m^2/4 = \rho_{DE}$ gives n ≈ 127. The matter chain is unchanged with μ = 1 (4.3× vs 4.4×). PURE-TWIST also predicts slower growth, $f\sigma_8$ about 3% below ΛCDM at z = 0.3–1.5, E_G about 5% higher, and $S_8 = 0.793$ (vs 0.798) for equal primordial amplitude; these are frozen before DESI DR3 full-shape.

Newer Planck data (V6, pre-registered). Refitting FLAG-1 and ΛCDM with Planck PR4 (NPIPE CamSpec high-ℓ) in place of PR3, together with DESI DR2 BAO, Pantheon+ and DES Y6 S_8, gives Δχ² = −7.81 for FLAG-1 (−3.91 with PR3); the locked rule reads this as HOLDS ON PR4. Best fits: FLAG-1 $H_0 = 67.0$, $\Omega_m = 0.314$, $\sigma_8 = 0.802$, β = 0.040; ΛCDM H_0 = 68.0, $\Omega_m = 0.305$. On the same data with n_s free (NS-PURE), PURE-TWIST fits better than ΛCDM by Δχ² = −3.85 with no extra parameters (H_0 66.8, Ω_m 0.313, σ_8 0.786). The locked reading is NO VERDICT: its gate needed ΛCDM's n_s within 0.005 of 0.969 and found 0.9635. That value lies inside the Twist's inflation prediction (0.963–0.965), so on Planck PR4 + DESI the n_s tension of Section 3 is absent; it appears only in ACT/SPT-combined analyses. PURE-TWIST itself prefers n_s 0.9677, so the late-time Twist does not lower n_s.

Corrections found on 13 Nov. (i) With the original cliff factor the field freezes at $x \approx 360$ after kination, where the cliff's leftover slope is $1.7\times\sqrt{(2/3)}$; TAIL-F (locked) rejects that model ($\Delta\chi^2$ +29.2) and, with the heat share freed, drives it to the PURE-TWIST limit (0.01%, $\Delta\chi^2$ −2.79). The revised cliff (Section 2) gives that limit at the computed heat share. (ii) The kination phase after the cliff adds about 2 e-folds: $N^* \approx 57.6$, $n_s \approx 0.9654$, $r \approx 0.0035$; supersymmetric loop corrections from the reheating field add about +0.002 (estimate). These are upward systematics on the locked n_s range (0.963–0.965), found after locking; the locked rule is scored as written. (iii) Kination reduces gravity-only dark-matter production from the bath, raising the required fermion mass to ≈ $1–6\times10^8$ GeV (estimate).

6. Consistency tests and negative results

The model of Section 2 is the survivor of a sequence of failures; each one ruled out a simpler version and fixed a feature of the final one.

Version tested	Result	What it forced
Today's slowly rolling field as the matter source	Too slow by ~10^{51}	Matter must form right after inflation
Straight-line tilt along the angle	Knife-edge: a 1% change in tilt moves r from 0.022 to 0.030; a little more ends inflation	No linear tilt
Fixed periodic potential	The angle settles into its minimum during inflation; r = 0	The minimum must move (a kick)
E-model with a vacuum, F = T − ½ (+ kick)	The kick works ($r \approx 0.16\,\delta$), but at the vacuum the angle has the inflaton's mass, ~3×10^{13} GeV, and oscillates 3–30,000 times per expansion time, erasing the asymmetry	No vacuum: the potential must run into a tail (this holds for any holomorphic F with a simple zero)
Plateau joined smoothly to a $\sqrt{(2/3)}$ tail	Inflation never ends ($\varepsilon \to 1/3$)	A cliff is needed

Version tested	Result	What it forced
Plateau → cliff → tail, inflaton falling toward small T	The angle becomes light ($m/H \approx 0.04$) but the curved space drains its speed by ~e^{-2} per e-fold	Mirror the model ($T \to 1/T$)
Mirrored, fields only	The speed is boosted; with no radiation the angle reaches 97% of the energy in ~3 e-folds	Include radiation from reheating
Mirrored, tail before the cliff	The tail inflates on its own (+7 e-folds), $n_s = 0.961$	Put the tail after the cliff (Section 2)

Cosmic birefringence. ACT + Planck report a rotation of CMB polarization of $0.277° \pm 0.057°$ (4.8σ). The Twist couples only through a derivative coupling to fermion spin, and for fermions far heavier than CMB photons such a coupling decouples (Quevillon & Smith 2019), so the Twist predicts essentially zero. Even if the full anomaly term survived, the field's measured roll since recombination ($\Delta\varphi = 0.11–0.29\ M_P$) gives at most 0.07°–0.19°. If the signal is cosmological, something other than this field produces it. This check was made after the measurement was known.

We regard this as one of the theory's sharpest falsification points and locked it (PREREG_BIREF_NULL, 13 Nov 2026 01:27 CT): the first future measurement with a polarisation-angle calibration independent of the Galactic-foreground assumption and total error $\leqslant 0.10°$ is scored SUPPORTED if $|\beta| < 2\sigma$ and NOT SUPPORTED if $\beta \geqslant 5\sigma$. Letting the Nieh–Yan anomaly survive gives at most 0.075° (FLAG-1 with DESI) and is recorded as the only fallback.

Gravitino. Reheating at $\geqslant 10^{13}$ GeV would normally overproduce gravitinos. In the no-scale setting with supersymmetry broken above the inflaton mass (Dudas, Gherghetta, Mambrini & Olive 2017), a gravitino heavier than T_{RH} is simply not produced. But the gravitino cannot then be the dark matter: that needs $T_{RH} \lesssim 10^{12}$ GeV (Dudas, Mambrini & Olive 2017, Eq. 9), while matter formation here needs $\geqslant 10^{13}$ GeV.

Dark-matter coupling. In this supergravity class, matter masses do not depend on the inflaton (Akrami et al. 2017), so the natural coupling of dark matter to the field is $\beta = 0$. A modular-weight coupling would give $\beta \approx 0.41k$; the fitted $\beta \approx 0.03–0.04$ would need $k \approx 0.08$, which is unnatural. This motivates the PURE-TWIST test (Section 5).

Gravity-only dark matter (exploratory). Dark matter coupled to the Standard Model only through gravity is produced in the plasma by graviton exchange at rate $R = \beta\, T^8/M_P^4$ [Garny, Sandora & Sloth, arXiv:1511.03278; Cléry, Mambrini, Olive & Verner, arXiv:2112.15214]. At the Twist's reheating temperature it gives the observed abundance for a mass of 10^5–10^7 GeV (light branch, scalar or fermion), or about 10^6–10^7 GeV at the derived T_{RH}; the heavy branch (~10^{16} GeV) survives only for reheating efficiency below ~10%. Inflation itself would overproduce a light, minimally coupled scalar by 10^{22}–10^{25} (frozen fluctuations $3H^4/16\pi^2$), so the dark matter must be a fermion (or a conformally coupled scalar): a gravity-only fermion of 1.4–6.6×10^6 GeV at the derived T_{RH}. Prediction: null results in direct detection.

7. Frozen predictions

These were fixed on 12 November 2026, before any of the deciding data exist. Each can fail.

Quantity	Prediction	Deciding measurement	Fails if
Tensor-to-scalar ratio r	0.0037–0.0040 (pre-registered band; full calculation 0.0034–0.0040)	first r measurement with $\sigma(r) \leqslant 0.002$ (LiteBIRD, CMB-S4 class)	$r < 0.002$ or $r > 0.008$ at 95%
Spectral index n_s	0.963–0.965	next CMB release (Simons Observatory, SPT-3G full, ACT final)	the data exclude 0.963–0.965 at 3σ (criterion stated here; the locked rule only reports n_s)
Cosmic birefringence from this field	≈ 0	CMB polarization with controlled angle calibration	a confirmed cosmological rotation that needs this field (another source could explain it)
Late dark energy	exponential tail of slope $\sqrt{(2/3)}$; dark matter uncoupled ($\beta = 0$)	DESI DR3, Euclid, LSST	the data require $w(z)$ a $\sqrt{(2/3)}$ tail can't produce

Quantity	Prediction	Deciding measurement	Fails if
Growth of structure (flagship, $\beta \approx 0.041$)	E_G 3–5% above ΛCDM at $0.3 < z < 1.5$ (e.g. 0.4209 vs 0.4015 at $z = 0.5$)	lensing × velocity surveys reaching ±2.4%	E_G consistent with ΛCDM at that precision

The E_G prediction belongs to the flagship ($\beta = 0.041$). If PURE-TWIST ($\beta = 0$) is preferred, E_G would instead match ΛCDM; both are recorded so that neither can be chosen after the fact.

PURE-TWIST at its best fit, for the next BAO release. The field thaws: $w = -0.899$ today, rising from −0.996 at $z = 3$ (CPL fit for $z < 2$: $w_0 = -0.900$, $w_a = -0.143$). Compared with ΛCDM at its own best fit, with neither refitted:

z	D_M/r_d vs ΛCDM	D_H/r_d vs ΛCDM
0.30	+0.86%	−0.06%
0.51	+0.40%	−0.55%
0.71	+0.16%	−0.66%
0.93	0.00%	−0.63%
1.32	−0.12%	−0.45%
2.33	−0.16%	−0.10%

The signature is expansion 0.45–0.66% faster than ΛCDM at $z = 0.5$–1.3, the precision the next DESI release should reach. A scoring rule for that release (both models frozen, no refit; SUPPORTED if $\Delta\chi^2 \leqslant -2$, DISFAVOURED if $\geqslant +2$) is drafted and awaits locking.

Further locked predictions (13 November 2026). Four more rules were hashed before any data that will judge them:

Rule	Prediction	Fails if
WA-CEILING	every thawing Twist version gives an effective $w_a \geq -0.40$ (FLAG-1 −0.27, PURE-TWIST −0.10; the dark-matter coupling makes w_{eff} appear to cross −1, but saturates near −0.30)	DESI DR3 headline $w_a + 3\sigma < -0.40$
BIREF-NULL	isotropic birefringence $\beta = 0$	a calibration-independent $\beta \geq 5\sigma$ (error $\leq 0.10°$)
LAB-NULLS	no fifth force, $\eta = 0$, no α drift, $c_{gw} = c$; a static cosmic-frame spin coupling of $2.8–4.0\times10^{-34}$ eV	a confirmed detection in any class
PURE-MCMC	posteriors for PURE-TWIST and ΛCDM on the tournament likelihood; consistency check of the best fits	(precision check; running)

For context, today's DESI DR2 + CMB + DES Y5 value $w_a = -0.73$ (+0.21) is a tension under WA-CEILING's readings, and the ACT + Planck birefringence $\beta = 0.277° \pm 0.057°$ is a 4.9σ tension with BIREF-NULL; neither is scored, because neither is the data the rules name. The hidden axion's inflationary fluctuations give a baryon isocurvature fraction $\beta_{iso} \approx 3\times10^{-4}$, far below Planck's limit of about 0.04.

8. Relation to prior work

Every mechanism in the chain has been published before; what we have not found in the literature is their combination in one field with a derived coupling.

Ingredient	Prior work
A rotating field drives leptogenesis through the Weinberg operator, then sphalerons	Lepto-axiogenesis (Co, Fernandez, Ghalsasi, Hall & Harigaya, JHEP 2021); axiogenesis (Co & Harigaya)
A rolling pseudoscalar as a chemical potential for baryon or lepton number	Spontaneous baryogenesis (Cohen & Kaplan 1987)
Kination after inflation sets an axion rotating, which then makes matter	Flipped rotating axion (arXiv:2502.08720)

Ingredient	Prior work
Axion kination and its gravitational-wave and CMB signals	Co et al. (arXiv:2108.09299)
Plateau plus dark-energy tail in no-scale supergravity	Akrami, Kallosh, Linde & Vardanyan (arXiv:1712.09693); Dimopoulos & Owen (arXiv:1703.00305)
The Immirzi field as a source of the baryon asymmetry	Mir, Shabir, Tiwari & Faizal (arXiv:2604.03973; abstract only was read)
Immirzi field coupled to fermions through Nieh–Yan	Mercuri (2009); Calcagni & Mercuri (2009)
High-scale supersymmetry with a superheavy gravitino in no-scale inflation	Dudas, Gherghetta, Mambrini & Olive (arXiv:1710.07341); Dudas, Mambrini & Olive (arXiv:1704.03008)

The proposed contribution is narrow:

1. A superpotential sector for the Gates–Ketov–Yunes Barbero–Immirzi superfield (whose $\alpha = 1$ geometry was the starting point) that gives both the inflaton potential and an angle dependence, which a constant superpotential cannot.
2. Using that partner, with the derived coupling $\sqrt{(3/8)}$, for lepto-axiogenesis, with the kick at the cliff and the curved-space boost during kination.
3. Placing it in one plateau → cliff → tail potential whose tail slope ($\sqrt{(2/3)}$) is tested against late-time data.
4. Testing all of it under pre-registered rules, negative results included.

Mir et al. use a derivative coupling to the B−L current and a gravitational Chern–Simons term, and predict birefringence and chiral gravitational waves; the Twist predicts no birefringence from this field, which separates the two.

9. Limitations

The main limits are that key pieces are assumed rather than derived, and that the late-time evidence is weak.

- **Kick term.** The kick cannot come from the dilaton's own non-perturbative sector (Section 4). A sequestered hidden field works, but it is an added ingredient, whose mass (~10^{10} GeV) is fixed by the observed matter, not derived; the form of the hidden

sector is assumed. Its phase γ sets the sign of the asymmetry, which is therefore not predicted.

- **Reheating.** Instant preheating is put in by hand with efficiency f; the coupling and decay chain that would produce it are not modelled.
- **Cliff.** Numerics use n = 10; the realistic drop to today's dark energy needs n ≈ 125. Inflation was checked at n = 125, but the post-inflation history was not.
- **Thermal treatment.** The rate equation uses one set of equilibrium relations (valid above 10^{12} GeV), stops at 10^{11} GeV, and takes the radiation from instant preheating as thermal at once.
- **Two descriptions of the late universe.** The late-time data tests use a phenomenological scalar with an exponential potential and coupling β (the flagship) or its β = 0 limit. The supergravity potential's tail was not itself fitted to data.
- **Fine-tuning of today's dark energy.** The cliff must drop the energy by ~10^{110}, as in every quintessential-inflation model.
- **Statistics.** Against ΛCDM the flagship's preference is 2.2–2.8σ locally and 0.2–1.0σ after correcting for the number of tests. When DESI is withheld, the flagship predicts DESI worse than ΛCDM.
- **Literature.** Two related papers (arXiv:2604.03973 and 1710.07341) were read in abstract only.
- **Heavy superpartners.** At the derived $T_{RH} \approx 3–5\times10^{14}$ GeV, superpartners near $m_{3/2} \approx 10^{11}–10^{13}$ GeV are produced; the model assumes R-parity violation so the lightest decays. Order-of-magnitude estimates leave a wide safe window (couplings $10^{-5}–10^{-3}$: no washout of the asymmetry, no relic), but this is an added assumption.
- **n_s and w_a.** Two measurements press on the theory: n_s (CMB + DESI ~0.973 vs predicted 0.963–0.965) and w_a (DESI DR2 + DES Y5 ~−0.73 vs the Twist's ceiling −0.40). Both are locked as tests rather than adjusted.

Dark matter is not explained. The data prefer it uncoupled from the field (β = 0), and it is not the gravitino, the Immirzi angle or the hidden field σ.

Further limitations (13 Nov). The helper field S is tachyonic without the stabilising term ($m_S^2 \approx -2H^2$). The main open problem is a fifth force: if Re T is today's dark energy, ordinary matter feels it. Through the Kaplunovsky–Louis anomaly terms the QCD scale moves with x for every matter modular weight except 3/4 (sequestered matter: $\Lambda_{QCD} \propto x^{-1/2}$, coupling β ≈ 0.41, about 120 times the Cassini bound), and the weight-3/4 case leaves quark masses moving (β ~ 0.015). No standard screening mechanism (chameleon, symmetron, least coupling, Vainshtein) applies to an exponential tail. Fixes that park Re T

in a minimum today meet the cosmological moduli problem unless the minimum is reached right after the cliff, which gives up the light tail; we keep the light tail and state this problem openly. Allowing each MSSM multiplet its own modular weight does not help: the conditions for fixed gauge couplings, Yukawas and μ have no common solution. Our late-time fits treat ordinary matter as uncoupled, so they test the tail's shape given a working shield, not the unshielded theory. A candidate resolution (estimate): the coupling comes from placing the Standard Model inside the no-scale supergravity, where Re T acts as a conformal factor. If instead the Standard Model is not supersymmetric and couples minimally to the Einstein-frame metric, meeting the twist field only through gravity, the Nieh–Yan term and the reheating field χ, the tree-level force vanishes and the surviving χ-loop coupling passes Cassini and MICROSCOPE for a χ–Standard-Model Yukawa $h \lesssim 0.05$, provided the Standard Model's coupling to χ is cut off at or below the GUT scale. Quantum stability of the tail requires every particle heavier than ~6 meV to have a T-independent mass (the Kolda–Lyth problem); this is assumed, not shown. The field travels $\Delta\varphi \approx 20\ M_P$ from the CMB point to today, in tension with the swampland distance conjecture. The dark-energy scale ($n \approx 124$) is chosen. The link to loop quantum gravity is the origin of the superfield only: today the angle sits at zero, so the field does not take LQG's black-hole value $\gamma \approx 0.24$ under the mapping $1/\gamma = y/x$. The cliff factor was revised after a failed locked test.

10. Methods, data and reproducibility

All code, pre-registered rules and results are available from the author on request (AEGIS repository, branch `unify-bg`).

Late-time likelihood. Every TOURNAMENT-family test uses one compressed likelihood:

- Pantheon+ supernovae, with the absolute magnitude marginalized analytically;
- CMB distance priors ($100\theta^*$, ω_b, and $\omega_b + \omega_c$ at $z = 1090$);
- BAO from DESI DR2, or SDSS DR12 ($z = 0.38, 0.51$) with DR16 LRG and QSO.

Each model is fitted three ways (FULL with DESI; NOBAO, then predicting DESI and SDSS; and SDSS in place of DESI) with two Nelder–Mead starts. A gate requires the stored ΛCDM and FLAG-1 values to reproduce within 0.05 in χ^2. Boltzmann code: AxiCLASS with a coupled-quintessence extension, modified for redshift-binned couplings and validated against the unmodified build (background to 10^{-7}, $P(k)$ and σ_8 to 2×10^{-6}).

Theory calculations. Two-field background equations in e-fold time on the curved field space, with Christoffel terms, solved with LSODA (rtol 10^{-8}–10^{-10}). Radiation is added as a fluid after instant preheating. The baryon asymmetry uses a single-flavour rate equation for B−L with the thermally averaged Weinberg rate and sphaleron conversion 28/79. Slow-roll observables are evaluated exactly along the trajectory.

Pre-registration. Each test's rule (`field_tests/rules/PREREG_*.md`) carries a SHA-256 hash and a lock time recorded before any of its numbers were computed. Deviations are logged in `*_DEVIATIONS.md` before any rerun; every reading is reported once. Rules used here: TOURNAMENT, KAPPA-ALPHA1, PURE-TWIST, MATTER, TWIST-INFLATION, DESI-HOLDOUT, MISSING-PIECE, CONSISTENCY.

Script	What it computes
`field_tests/missing_piece/tournament.py`, `kappa_alpha1.py`, `pure_twist.py`	late-time fits and predictions
`theory/matter/boltzmann_leptogenesis.py`	MATTER
`theory/inflation/twist_inflation_N.py`	TWIST-INFLATION
`theory/inflation/sugra_mirror.py`, `sugra_mirror_radiation.py`, `theory/matter/chain_d22.py`	the model of Section 2 and its matter yield
`theory/inflation/sugra_kick.py`, `sugra_tail.py`, `sugra_cliff*.py`, `angle_restart_*.py`, `theory/matter/free_roll_test.py`, `birefringence_check.py`, `gravitino_check.py`	the tests of Section 6

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